

Network Position and Academic Performance

Adding Network Centrality to Smirnov & Thurner (2017)

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Presentation 2 — Progress Report

Does network position predict grades?

On top of who a student's direct friends are

Recap: Smirnov & Thurner (2017) show grades and friends' grades are linked mostly by *selection*, not influence — but they look at one number, a friend's average GPA. Calvó-Armengol, Patacchini & Zenou (2009) show that *position* (Katz–Bonacich centrality) predicts achievement — but only in a single snapshot, so cause and effect are not separated.

Our plan: take the centrality prediction to Smirnov & Thurner's panel data, and test it while holding friend-group GPA and a student's own past GPA fixed.

(Calvó-Armengol, Patacchini & Zenou 2009; Smirnov & Thurner 2017)

Recap: design and data

The empirical strategy from Presentation 1

$$\text{GPA}_{i,t+1} = \beta_0 + \beta_1 \text{Katz}_{i,t} + \beta_2 \text{GPA}_{i,t} + \beta_3 \overline{\text{GPA}}_{i,t}^{\text{friends}} \\ + \beta_4 \text{InDeg}_{i,t} + \beta_5 \text{OutDeg}_{i,t} + \gamma_{\text{group}} + \varepsilon_{i,t}$$

Each control has a job: friend GPA nets out friend-group composition; in- and out-degree net out raw popularity; ϕ is swept from near 0 (popularity) to near its ceiling (long reach).

The data: 5 cohorts, 655–1,549 students each, directed “like” ties every ~ 3 months.

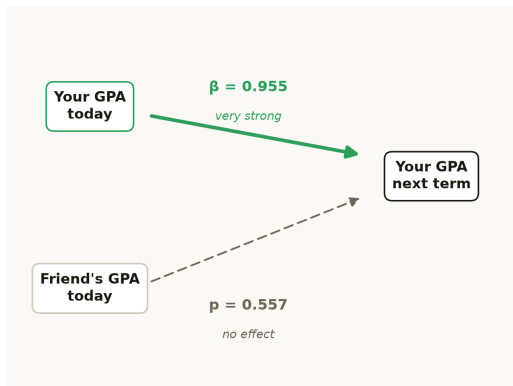
GPA repeats over time for the school group only — each university cohort has one static GPA value.

What we did

1. **Replicate** — confirm the data behaves the way Smirnov & Thurner reported.
 2. **Naive test** — centrality alone, no popularity control.
 3. **Planned test** — add popularity, sweep ϕ , pool all 5 groups.
 4. **Check & fix** — test the assumptions; found one had failed.
- Here's what actually happened when we ran it.

The replication holds

Own past GPA vs. a friend's past GPA, school group, $n = 2,088$



- New friendships are also more GPA-similar than dissolving ones (university $p < 0.001$; same direction in school, $p = 0.183$). Both match Smirnov & Thurner — the foundation the rest of this talk builds on.

The naive effect is popularity, relabelled

Katz-Bonacich alone vs. with a degree control, pooled, $n = 36,696$



- In-degree itself: $+0.153$, $p < 0.001$ — that's what was driving the naive result.

The headline test

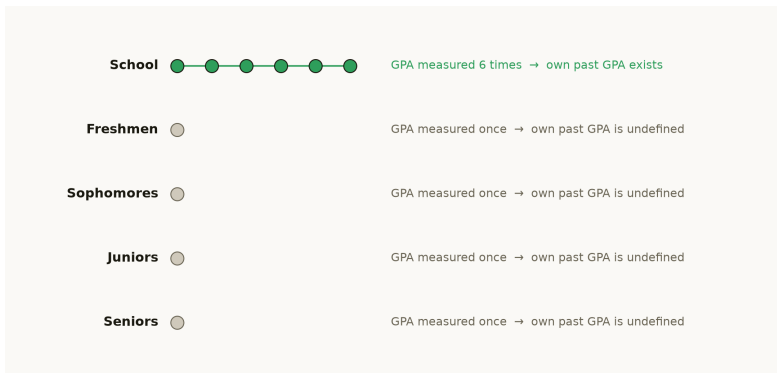
Net of popularity and friend GPA, pooled across all 5 groups

ϕ (frac. of ceiling)	0.05	0.20	0.40	0.60	0.80	0.95
Katz coefficient	+0.082	+0.022	-0.034	-0.051	-0.048	-0.036
p -value	0.012	0.598	0.407	0.115	0.031	0.011

- At $\phi = 0.85$, fixed in advance as the headline: $\beta = -0.0415$, $SE = 0.019$, $p = 0.033$, $n = 36,696$. Read at face value: position has a small, negative, independent effect.

But we found a gap in it

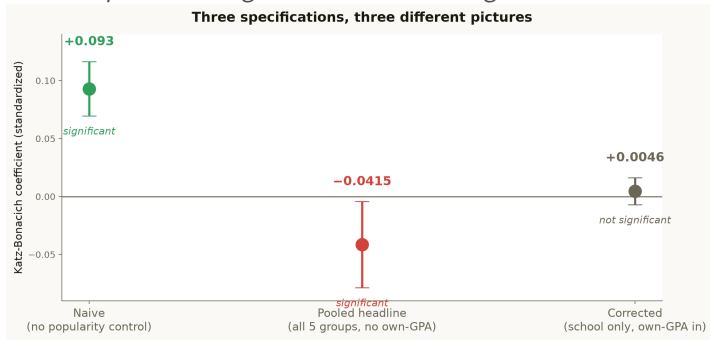
Own-past-GPA cannot be built as a regressor, pooled



- Presentation 1's model equation promised this control. The test we actually ran did not have it.

The corrected test

School only — the one sample where a genuine $t \rightarrow t+1$ design with an own-GPA control is possible



- Net of popularity, friend GPA, and a student's own past performance, we find no evidence that network position independently predicts grades ($n = 2,088$, 535 students — a much wider confidence interval than the pooled 36,696, but the only sample where the test is valid).

▸ Robustness

▸ Multicollinearity

▸ What's next

Conclusion

- **Net of popularity, friend GPA, and a student's own past performance, we find no evidence that network position independently predicts grades.**
- **Why this is still worth reporting:** Naive centrality looked positive, the pooled sweep looked negative, the corrected test looks like nothing.

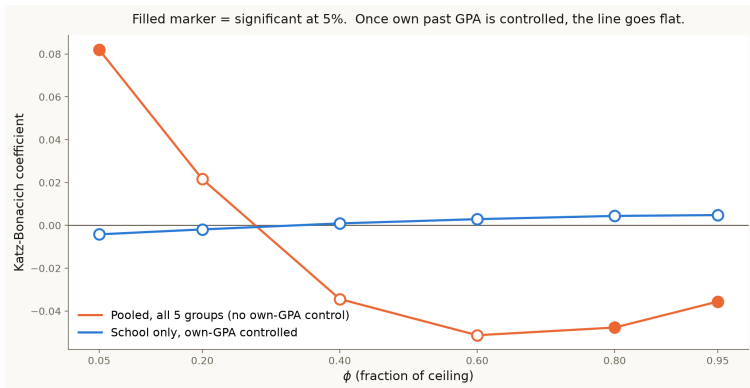
Implication: Interventions built around widening students' network reach are not supported by this data once the obvious confounds are handled. Next: testing whether weak, one-way "likes" are the reason we see no effect.

Appendix

Sample and network construction

- **Source:** Smirnov & Thurner (2017) replication data, Harvard Dataverse, CC0. ~6,200 Russian students.
- **Nodes:** 655 high-school students; 1,200–1,549 per university cohort (freshmen/sophomores/juniors/seniors).
- **Edges:** directed “like” ties, $i \rightarrow j$ if i liked j at least once in a ~3-month window. 2–14 snapshots per group, 38 networks total. Only 24–27% of ties go both ways.
- **GPA measurement:** 6 repeated measurements for school; 1 static measurement for each university cohort.

Robustness: the full ϕ range



- Before the fix, pooled swings from significantly positive to significantly negative. After it, school-only is flat across the whole range.

Checking for multicollinearity

Does adding own-past-GPA collide with the other regressors?

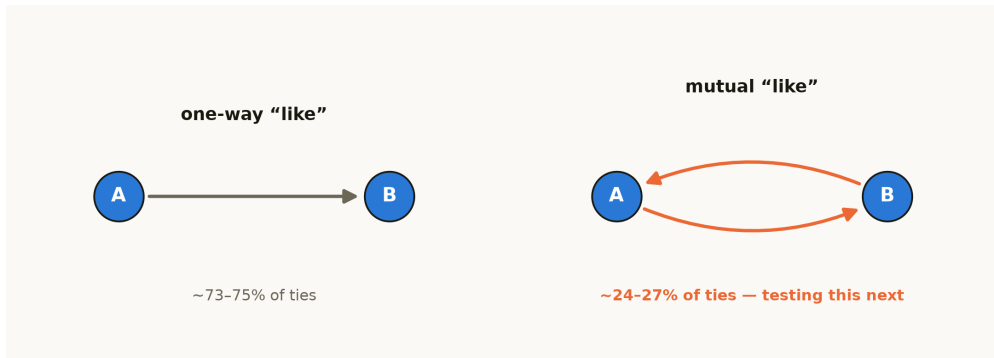
Own-past-GPA correlates weakly with the other right-hand-side variables:

Correlates with	Katz	In-degree	Out-degree	Friend GPA
Correlation	0.008	0.065	0.122	0.285

Variance inflation factor for own-past-GPA: **1.11** — no collinearity problem.

- Katz and in-degree still correlate at 0.866 ($VIF \approx 4$) — expected, and exactly what sweeping ϕ is for.

What's next: reciprocal ties



- Recompute centrality using only mutually-confirmed ties — testing directly whether weak, one-way “likes” are the reason we see no effect.

Related literature

- **Calvó-Armengol, Patacchini & Zenou (2009), IZA DP 3859:** the model we test, on US AddHealth data, one snapshot — cause and effect not separated.
- **Smirnov & Thurner (2017), PLOS ONE:** the paper and data we replicate and extend; their method looks at one number, a friend's average GPA.
- **Giulietti, Vlassopoulos & Zenou (2020), IZA DP 13680:** peer effects via reference groups, not friendship ties.

Positioning: this is the first time Katz–Bonacich centrality has been tested with a real time lag and an own-performance control — Calvó-Armengol et al.'s cross-sectional data could not support that.